

Hemoglobinopathy — The Cyanosis Courtroom

HYPOXEMIA



$\text{SpO}_2 \downarrow$
 $\text{PaO}_2 \downarrow$

Test: ABG

Tx: OXYGEN

M Hb



$\text{SpO}_2 \downarrow$
 $\text{PaO}_2 \downarrow$

Test: KCN-NO

Tx: NONE

MetHb



$\text{SpO}_2 \downarrow$
 $\text{PaO}_2 \downarrow$

Test: KCN-RED

Tx: MB 1-2

SulfHb



$\text{SpO}_2 \downarrow$
 $\text{PaO}_2 \downarrow$

Test: CO-OX

Tx: WAIT 120D

MASTER COMPARISON CHART

HYPOXEMIA	M Hb	MetHb	SulfHb
			
$\text{SpO}_2 \downarrow$ $\text{PaO}_2 \downarrow$	$\text{SpO}_2 \downarrow$ $\text{PaO}_2 \downarrow$	$\text{SpO}_2 \downarrow$ $\text{PaO}_2 \downarrow$	$\text{SpO}_2 \downarrow$ $\text{PaO}_2 \downarrow$
Test: ABG	Test: KCN-NO	Test: KCN-RED	Test: CO-OX
Tx: OXYGEN	Tx: NONE	Tx: MB 1-2	Tx: WAIT 120D

MetHb



$\text{SpO}_2 \downarrow$
 $\text{PaO}_2 \downarrow$

Test: KCN-RED

Tx: MB 1-2

SulfHb



$\text{SpO}_2 \downarrow$
 $\text{PaO}_2 \downarrow$

Test: CO-OX

Tx: WAIT 120D

COHb



$\text{SpO}_2 \downarrow$
 $\text{PaO}_2 \downarrow$

Test: CO-OX/ABG

Tx: 100% O2

LOW AFFINITY



$\text{SpO}_2 \downarrow$
 $\text{PaO}_2 \downarrow$

Test: P50/HPLC

Tx: NONE

MASTER COMPARISON CHART

MetHb	SulfHb	COHb	LOW AFFINITY
			
$\text{SpO}_2 \downarrow$ $\text{PaO}_2 \downarrow$	$\text{SpO}_2 \downarrow$ $\text{PaO}_2 \downarrow$	$\text{SpO}_2 \downarrow$ $\text{PaO}_2 \downarrow$	$\text{SpO}_2 \downarrow$ $\text{PaO}_2 \downarrow$
Test: KCN-RED	Test: CO-OX	Test: CO-OX/ABG	Test: P50/HPLC
Tx: MB 1-2	Tx: WAIT 120D	Tx: 100% O2	Tx: NONE

COHb



LEFT SHIFT Clutching O₂ tightly

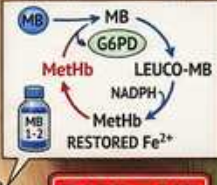
DOCK 1

HIGH AFFINITY

TRAP 1 FALSE NORMAL = COHb

MetHb




G6PD + MB = BAD
FALSE MetHb NO REDOSE

DOCK 2

TRAP 2 BROWN BLOOD = MetHb

SulfHb



PERMANENT SULFUR
120 DAYS
REMOVE DRUG WAIT

DOCK 3

TRAP 3 KCN-NO = M Hb

M Hb



TYPE I (RBC)
TYPE II (ALL CELLS) severe neurologic cases

DOCK 4

NO TX

TRAP 4 G6PD + MB = BAD

TRAP 5 MB FAILS CO-OX

TRAP 6 SulfHb + MB = NO

TRAP 7 HIGH AFFINITY = HbP₅₀

TRAP 8 Low Ferritin Masked

TRAP 9 Yersinia + Thal

TRAP 10 HbSS + α-Thal

1. Carboxyhemoglobin (COHb)

WHAT YOU SEE

Dock 1: cherry-red patient, fire/smoke, pulse ox falsely normal

WHAT IT ENCODES

Carbon monoxide binds hemoglobin with ~200–250× the affinity of O₂, forming carboxyhemoglobin and also LEFT-shifting the dissociation curve of remaining heme, so O₂ is held tightly and not released — a double hit causing severe tissue hypoxia despite a normal PaO₂ (dissolved O₂ is unaffected). Classic source: smoke inhalation, faulty heaters, car exhaust. Skin is described as 'cherry-red' (often a late/postmortem sign). Symptoms: headache, confusion, syncope; check a COHb level by co-oximetry.

BOARD TRAP

WRONG: 'A normal pulse oximetry reading rules out CO poisoning.' Standard 2-wavelength pulse oximetry cannot distinguish COHb from O₂Hb, so SpO₂ reads FALSELY NORMAL — you must measure COHb with co-oximetry/ABG. PaO₂ is also normal and misleading.

2. COHb treatment

WHAT YOU SEE

100% O₂ mask, hyperbaric chamber

WHAT IT ENCODES

Treatment of CO poisoning is high-flow 100% O₂ by non-rebreather, which competitively displaces CO and shortens the COHb half-life from ~4–6 hours (room air) to ~60–90 minutes. Hyperbaric O₂ (HBO) further shortens it to ~20–30 minutes and is considered for severe cases: loss of consciousness, neurologic deficits, COHb >25% (>15–20% in pregnancy), pregnancy with fetal distress, ischemic ECG/troponin, or severe acidosis — aiming to reduce delayed neurocognitive sequelae.

BOARD TRAP

WRONG: 'Methylene blue treats carbon monoxide poisoning.' Methylene blue is for methemoglobinemia, not COHb; CO is treated with 100% O₂ ± hyperbaric oxygen.

3. Methemoglobin (MetHb)

WHAT YOU SEE

Dock 2: chocolate-brown blood, oxidant drugs

WHAT IT ENCODES

Methemoglobin is hemoglobin with iron oxidized to the ferric Fe³⁺ state, which cannot bind O₂; worse, it left-shifts the remaining ferrous heme, impairing O₂ release → functional anemia/tissue hypoxia despite normal PaO₂. Blood is classically chocolate-brown and does not turn red on exposure to air/O₂. The hallmark is a 'saturation gap' (normal PaO₂ but low SaO₂ by co-oximetry) with SpO₂ characteristically stuck around ~85% regardless of supplemental O₂. Symptoms appear above ~15–20% MetHb; >50–70% can be lethal.

BOARD TRAP

WRONG: 'A low SpO₂ that won't rise with 100% O₂ means a shunt/severe lung disease.' A pulse ox plateauing near 85% with a NORMAL PaO₂ (saturation gap) points to methemoglobinemia — confirm with co-oximetry, not a chest CT.

4. MetHb causes

WHAT YOU SEE

Dapsone, benzocaine, nitrites, nitrates listed

WHAT IT ENCODES

Acquired methemoglobinemia is driven by oxidant exposures: dapsone (a top board cause), topical/local anesthetics (benzocaine spray, prilocaine/EMLA, lidocaine), nitrites and nitrates (amyl nitrite 'poppers', well-water nitrates in infants, nitroglycerin/nitroprusside), sulfonamides, and aniline dyes. Infants are especially susceptible (lower NADH-cytochrome b₅ reductase activity). Congenital forms: cytochrome b₅ reductase deficiency (autosomal recessive enzymatic) and Hb M (autosomal dominant globin variant).

BOARD TRAP

WRONG: 'Cyanosis in a baby on well water is congenital cyanotic heart disease.' Nitrate-contaminated well water causes acquired methemoglobinemia ('blue baby' from nitrates) — look for chocolate blood and a saturation gap before invoking a cardiac shunt.

5. Methylene blue treatment

WHAT YOU SEE

Methylene blue + NADPH restores Fe²⁺

WHAT IT ENCODES

Methylene blue (1–2 mg/kg IV) is first-line for symptomatic methemoglobinemia. It is reduced by NADPH-methemoglobin reductase (using NADPH generated by G6PD) to leukomethylene blue, which then non-enzymatically reduces Fe³⁺ back to functional Fe²⁺. Onset is within 30–60 minutes; redose if needed. Treat when MetHb is symptomatic or roughly >20–30% (or lower with cardiopulmonary disease).

BOARD TRAP

WRONG: 'Methylene blue directly donates electrons without needing NADPH/G6PD.' It works THROUGH the NADPH-dependent pathway (G6PD-supplied) — which is exactly why it fails and is dangerous in G6PD deficiency (next panel).

6. MetHb + G6PD danger

WHAT YOU SEE

Red box: G6PD + MB = BAD

WHAT IT ENCODES

In G6PD deficiency methylene blue is BOTH ineffective and dangerous: without adequate NADPH, MB cannot be reduced to its active leukomethylene-blue form (so it doesn't reverse MetHb), and MB itself acts as an oxidant that can precipitate acute hemolysis. Use alternatives instead: high-dose ascorbic acid (vitamin C), and for severe/refractory cases exchange transfusion or hyperbaric O₂.

BOARD TRAP

WRONG: 'Methylene blue is the universal antidote for methemoglobinemia in any patient.' In G6PD deficiency it is contraindicated (ineffective + can cause hemolysis) — switch to ascorbic acid or exchange transfusion.

7. Sulfhemoglobin (SulfHb)

WHAT YOU SEE

Dock 3: greenish blood, sulfur, permanent

WHAT IT ENCODES

Sulfhemoglobin forms when a sulfur atom is incorporated into the heme ring (often from the same oxidant drugs — sulfonamides, dapsons, metoclopramide, phenazopyridine). The bond is IRREVERSIBLE: the affected RBC stays abnormal for its entire ~120-day lifespan. Blood is greenish/dark, cyanosis is out of proportion to the measured level (even small amounts cause visible cyanosis), and co-oximetry may overestimate MetHb unless specifically corrected.

BOARD TRAP

WRONG: 'Greenish or dark blood with cyanosis is just methemoglobinemia.' Sulfhemoglobinemia (greenish blood) is a separate, IRREVERSIBLE dyshemoglobin — and unlike MetHb it does not respond to methylene blue.

8. SulfHb management

WHAT YOU SEE

Remove drug, WAIT, MB doesn't work

WHAT IT ENCODES

There is NO antidote for sulfhemoglobinemia. Management is to stop the offending oxidant and WAIT for normal red-cell turnover (~120 days) as new, unaffected RBCs replace the abnormal ones; transfusion can be used if the patient is symptomatic. Methylene blue is useless because the sulfur-heme modification is covalent/irreversible (contrast with MetHb, where MB works).

BOARD TRAP

WRONG: 'Methylene blue reverses sulfhemoglobinemia like it does methemoglobinemia.' SulfHb is irreversible — MB does not work; you remove the drug and wait out the RBC lifespan.

9. Hb M (congenital)

WHAT YOU SEE

Dock 4: grey patient, pedigree, 'NO TX'

WHAT IT ENCODES

Hb M is inherited (autosomal DOMINANT) methemoglobinemia from a globin mutation in the heme pocket that locks iron in the Fe³⁺ state. It presents as lifelong, well-tolerated cyanosis (grey-blue) with a positive family history (pedigree) and is typically asymptomatic — patients are not in distress despite looking cyanotic. Because the defect is structural (globin), it does NOT respond to methylene blue and generally needs NO treatment.

BOARD TRAP

WRONG: 'Congenital cyanosis from Hb M should be treated urgently with methylene blue.' Hb M is a structural globin defect (not enzymatic), so MB is ineffective; it is well tolerated and requires no treatment — contrast with congenital cytochrome b5 reductase deficiency, which CAN respond to MB/ascorbic acid.

10. Master comparison chart

WHAT YOU SEE

Judge's chart: COHb / MetHb / SulfHb spectra and tests

WHAT IT ENCODES

Unifying frame: COHb, MetHb, and SulfHb are dyshemoglobins that cause cyanosis/hypoxia while ABG PaO₂ stays NORMAL (PaO₂ reflects only dissolved O₂, which these don't affect). Co-oximetry (multi-wavelength) is the key test that quantifies and distinguishes each species — ordinary 2-wavelength pulse oximetry is fooled (falsely normal in CO, plateaus ~85% in MetHb). Calculated SaO₂ from PaO₂ looks fine, producing a saturation gap.

BOARD TRAP

WRONG: 'A normal ABG PaO₂ means oxygen delivery is adequate.' PaO₂ is normal in all dyshemoglobinemias yet tissue O₂ delivery is impaired — you must measure functional saturation with co-oximetry, not infer it from PaO₂.

11. Trap: brown blood = MetHb

WHAT YOU SEE

Trap card: brown blood = MetHb

WHAT IT ENCODES

Board pearl: chocolate-brown blood that does NOT turn red on exposure to air/100% O₂, plus cyanosis unresponsive to oxygen and a saturation gap, equals methemoglobinemia until proven otherwise. Look for an oxidant trigger (dapsone, benzocaine, nitrites). Confirm with co-oximetry and treat with methylene blue (unless G6PD-deficient).

BOARD TRAP

WRONG: 'Brown blood means it's old/venous and clinically meaningless.' Fresh chocolate-brown arterial blood that won't redden with O₂ is a specific clue to methemoglobinemia — act on it.

12. Trap: false-normal SpO₂ = COHb

WHAT YOU SEE

Trap card: false normal SpO₂ = COHb

WHAT IT ENCODES

Board pearl: a falsely NORMAL pulse-ox reading in a patient with headache, nausea, confusion, or a fire/smoke/heater exposure points to carbon monoxide poisoning — standard pulse oximetry cannot see COHb. Confirm with carboxyhemoglobin level (co-oximetry); treat with 100% O₂ ± hyperbaric. Whole families/co-exposed individuals symptomatic together is a tell.

BOARD TRAP

WRONG: 'SpO₂ of 99% excludes hypoxia, so CO poisoning is off the table.' Pulse oximetry reads COHb as if it were O₂Hb (falsely high) — a normal SpO₂ does NOT rule out CO; measure COHb.

13. Trap: high-affinity → ↑Hb

WHAT YOU SEE

Trap card: high affinity = HbP50 low → erythrocytosis

WHAT IT ENCODES

Board pearl: unexplained erythrocytosis (high Hb/Hct) with a LOW P50, NORMAL SaO₂, and normal/high EPO but NO JAK2 mutation suggests a high-O₂-affinity hemoglobin variant. The left-shifted curve withholds O₂ from tissues, driving compensatory EPO and red-cell production. It is benign and compensatory — do NOT phlebotomize as you would polycythemia vera.

BOARD TRAP

WRONG: 'Erythrocytosis with normal SaO₂ must be polycythemia vera — start phlebotomy and check JAK2.' High-affinity Hb (low P50, normal SaO₂, no JAK2) is compensatory; phlebotomy worsens tissue O₂ delivery.

14. Traps: thal mimics

WHAT YOU SEE

Trap cards 9-10: Yersinia + thal, HbSS + α-thal

WHAT IT ENCODES

Two co-inheritance pearls: (1) Iron overload (transfusion-dependent thalassemia, hemochromatosis) plus deferoxamine raises the risk of Yersinia enterocolitica infection (Yersinia is siderophilic and uses deferoxamine as a siderophore) — suspect with febrile diarrhea/sepsis. (2) Co-inherited α-thalassemia in sickle cell disease (HbSS + α-thal) reduces MCHC and lowers MCV, mildly ameliorating the sickling/hemolysis but increasing retinopathy — a microcytic SCD patient may have concurrent α-thal or iron deficiency.

BOARD TRAP

WRONG: 'A sickle cell patient with microcytosis just has iron deficiency.' Co-inherited α-thalassemia also lowers MCV in SCD (and tempers hemolysis) — don't assume iron deficiency without checking iron studies and considering α-thal.